🛰️ Fraud GPS: Powered by Fraud DNA

An Interactive Cyber-Financial Investigation Platform

> **"Google Maps for stolen money, powered by graph intelligence."**

Table of Contents

1. Problem & Opportunity
2. Solution Overview
3. Core Concept — How It Works
4. The Two Engines
5. System Architecture
6. Data Flow Pipeline
7. Graph Schema & Data Model
8. Risk Scoring Framework
9. Explainability Engine
10. UI / UX Architecture
11. Tech Stack
12. Demo Storyline
13. MVP vs Advanced Feature Matrix
14. Hackathon Build Plan

1. Problem & Opportunity

The Crisis

India reported **31.2 lakh+ cybercrime complaints** on NCRP in 2023-24 alone. Bihar, adjacent to the infamous Jamtara belt (Jharkhand), is one of the worst-affected states. The dominant fraud pattern:

```
Victim is tricked (phishing / digital arrest / fake KYC / loan scam)
    ↓
Money is sent via UPI / IMPS to a mule account
    ↓
Money is rapidly SPLIT across 3-7 accounts within minutes
    ↓
Each split is LAYERED through 2-5 more hops
    ↓
Money is CASHED OUT via ATM withdrawal, crypto purchase, or merchant payment
    ↓
Trail goes cold — investigators are left chasing paper
```

Why Current Systems Fail

Current Approach	Why It Fails
Threshold-based alerts	Mules stay under ₹50K per transaction to dodge triggers
Manual complaint analysis	One officer can manually trace ~2-3 money trails per day
Siloed complaint databases	Case #142 and Case #987 involve the same mule network, but nobody knows
Static dashboards	No way to visually "follow the money" or see network structure
No cross-case intelligence	Each complaint is investigated independently — syndicates remain invisible

The Opportunity

> What if a cyber police officer could **paste a complaint**, hit "Trace", and **watch the stolen money flow across accounts on screen like a GPS route** — while the system automatically reveals that this fraud is connected to 47 other complaints through a shared mule network?

That's **Fraud GPS**.

2. Solution Overview

Fraud GPS: Powered by Fraud DNA is an interactive investigation platform with two intelligence engines and one cinematic UI layer:

Layer	Role	Analogy
Fraud GPS Engine	Traces where the money went	Google Maps route
Fraud DNA Engine	Reveals why this fraud is connected to others	DNA fingerprint match
Investigation UI	Makes it visual, interactive, and demo-ready	Command center screen

3. Core Concept — How It Works

When a fraud complaint arrives, the platform executes this flow:

The Investigation Experience (Step by Step)

Step	What Happens	What the Officer Sees
1	Complaint is loaded	Victim node appears (blue glow)
2	First transaction traced	Animated pulse flows to first receiver
3	Receiver analyzed	Node turns yellow/red based on risk
4	Forward trace continues	Money stream splits into branches (animated)
5	Layering detected	Chain of rapid hops lights up sequentially
6	Cash-out found	Terminal node pulses dark (ATM withdrawal)
7	Cross-case links found	Dashed lines connect to other complaint nodes
8	Fraud DNA match	Panel shows: "87% match with Campaign #Gamma"
9	Mule roles labeled	Nodes labeled: Collector, Splitter, Hub, Cash-out
10	Report generated	One-click PDF with evidence chain

4. The Two Engines

🛰️ Engine A: Fraud GPS — Money Trail Tracer

This is the **hero feature** — the visual investigation experience.

What it does:

```
Input:  Victim account + suspect transaction
Output: Complete money trail graph with splits, merges,
        layering chains, and cash-out points
```

Core Algorithms:

Algorithm	Purpose
Multi-hop BFS/DFS	Trace money forward from victim account through N hops
Split Detection	Identify where one incoming transfer fans out to multiple recipients
Merge Detection	Identify where multiple streams reconverge into one account
Rapid Transfer Detection	Flag accounts that forward >80% of received funds within 30 minutes
Layering Chain Detection	Identify sequential chains of pass-through accounts
Cash-out Terminal Detection	Identify endpoints: ATM withdrawals, merchant payments, crypto exchanges
Amount Decay Tracking	Track how the original amount diminishes through each hop (fees, partial withdrawals)

Visual Trail Behavior:

```
VICTIM (blue)
     │
     │  ₹50,000 ——— thick animated line
     │
COLLECTOR (orange)
     │
     ├——— ₹20,000 → MULE A (yellow) → ₹19,500 → CASH-OUT 1 (dark)
     │
     ├——— ₹15,000 → MULE B (red)    → ₹7,000  → MULE D → CASH-OUT 2
     │                                → ₹7,500  → MULE E → CASH-OUT 3
     │
     └——— ₹14,500 → MULE C (red)    → ₹14,000 → CASH-OUT 4 (dark)

Legend:
  Line thickness = amount
  Line brightness = speed of transfer
  Pulsing red edge = flagged suspicious
  Node color = risk level (green → yellow → orange → red → dark)
```

🧬 Engine B: Fraud DNA — Intelligence & Correlation

This is the **intelligence layer** that makes Fraud GPS more than just a pretty graph.

Four Types of Fraud DNA:

```
```

Entity DNA — Shared Identifiers

Cross-references these entity types across ALL complaints to find hidden connections:

Entity Type	Example Match
Bank Account	Same account appears in complaint #142 and #987
Phone Number	Same phone linked to 3 different mule accounts
Device ID	Same device used to operate 5 fraud accounts
IP Address	Same IP seen in login sessions of multiple mule accounts
UPI ID	Same UPI handle received money from 12 different victims
Email	Same email used for KYC of 4 accounts
Beneficiary Name	"Rahul Kumar" vs "RAHUL K" vs "R Kumar" — fuzzy match
IFSC / Branch	Cluster of mule accounts opened at same branch

Behavioral DNA — Pattern Signatures

Every fraud operation has a behavioral fingerprint:

Behavioral Signal	What It Reveals
Transaction timing pattern	Always operates between 10 PM - 2 AM
Split structure	Always splits into exactly 3 accounts
Average transaction amount	Stays between ₹8K-₹12K per transfer
Time-to-forward	Always forwards within 4-8 minutes
Number of hops	Always uses exactly 3 layers
Fan-out ratio	Incoming:Outgoing counterparties = 10:3
Cash-out method	Always ends in ATM withdrawal
Dormancy pattern	Account activated 2-3 days before fraud, dormant after

These signals are vectorized into a **Behavioral Fingerprint** per case. Similar fingerprints = likely same fraud syndicate.

Network DNA — Topology Similarity

Compares the **shape** of fraud networks:

```
Case A network:          Case B network:
   V1 → M1 → C1            V5 → M4 → C3
   V2 → M1 → C2            V6 → M4 → C4
   V3 → M2 → C1            V7 → M5 → C3
   V4 → M2 → C2            V8 → M5 → C4

Shape: 4-victim → 2-mule → 2-cashout (FAN-IN → SPLIT → FAN-OUT)

→ Network DNA similarity: 94% → SAME CAMPAIGN
```

Narrative DNA — Complaint Similarity

Signal	Example
Scam type	Both victims received "KYC expiry" SMS
Social engineering method	Both were told to install remote access app
Claimed platform	Both mentioned "SBI" in complaint
Call duration pattern	Both were on call for 45+ minutes
Language/script	Both complaints describe same script/dialogue

Fraud DNA Output Panel:

```
┌──────────────────────────────────────────────────────┐
│  🧬 FRAUD DNA ANALYSIS — Case #2024-BH-4521          │
│                                                      │
│  Fraud Signature Score: 0.89 / 1.00                  │
│  Campaign Family: "Gamma Network" (14 linked cases)  │
│                                                      │
│  ENTITY MATCHES:                                     │
│  ├─ Account XXXX4521 found in 6 other complaints     │
│  ├─ Phone +91-98XXXXX12 linked to 3 mule accounts    │
│  └─ Device ID matched across Case #142, #308, #987   │
│                                                      │
│  BEHAVIORAL MATCH:                                   │
│  ├─ Split pattern: 1→3 (matches Gamma signature)     │
│  ├─ Avg time-to-forward: 6.2 min (Gamma avg: 5.8 min)│
│  └─ Operating window: 11 PM - 1 AM (Gamma pattern)   │
│                                                      │
│  NETWORK MATCH:                                      │
│  ├─ Topology: FAN-IN → HUB → FAN-OUT                 │
│  └─ 94% structural similarity to Gamma cluster       │
│                                                      │
│  TOP LINKED CASES:                                   │
│  ├─ Case #142 — 92% DNA match                        │
│  ├─ Case #308 — 87% DNA match                        │
│  └─ Case #987 — 81% DNA match                        │
│                                                      │
│  SUSPECTED HUB: Account XXXX7734 (appears in 11 cases)│
│  SUSPECTED CONTROLLER: Phone +91-70XXXXX89           │
└──────────────────────────────────────────────────────┘
```

5. System Architecture

Full Architecture Diagram

Module Descriptions

Module	Responsibility	Key Input	Key Output
Data Parser	Normalize CSVs, JSONs into unified schema	Raw files	Cleaned records
Entity Extractor	Extract & normalize phones, UPIs, names	Transactions + complaints	Entity registry
Graph Builder	Build nodes (accounts) & edges (transactions)	Cleaned records	NetworkX graph
Trail Tracer	BFS/DFS forward trace from victim account	Start node + graph	Ordered trail path
Split Detector	Find 1→N fan-out points	Trail subgraph	Split events list
Entity DNA	Cross-reference entities across complaints	Entity registry	Match matrix
Behavioral DNA	Vectorize transaction patterns per case	Case transaction sets	Fingerprint vectors
Network DNA	Compare subgraph topologies	Case subgraphs	Similarity scores
Risk Scorer	Composite risk score per node	All engine outputs	Score + reasons
Role Classifier	Label node as Collector/Splitter/Hub/Cash-out	Graph features	Role labels
Explanation Generator	Human-readable suspicion reasons	Risk components	Explanation text

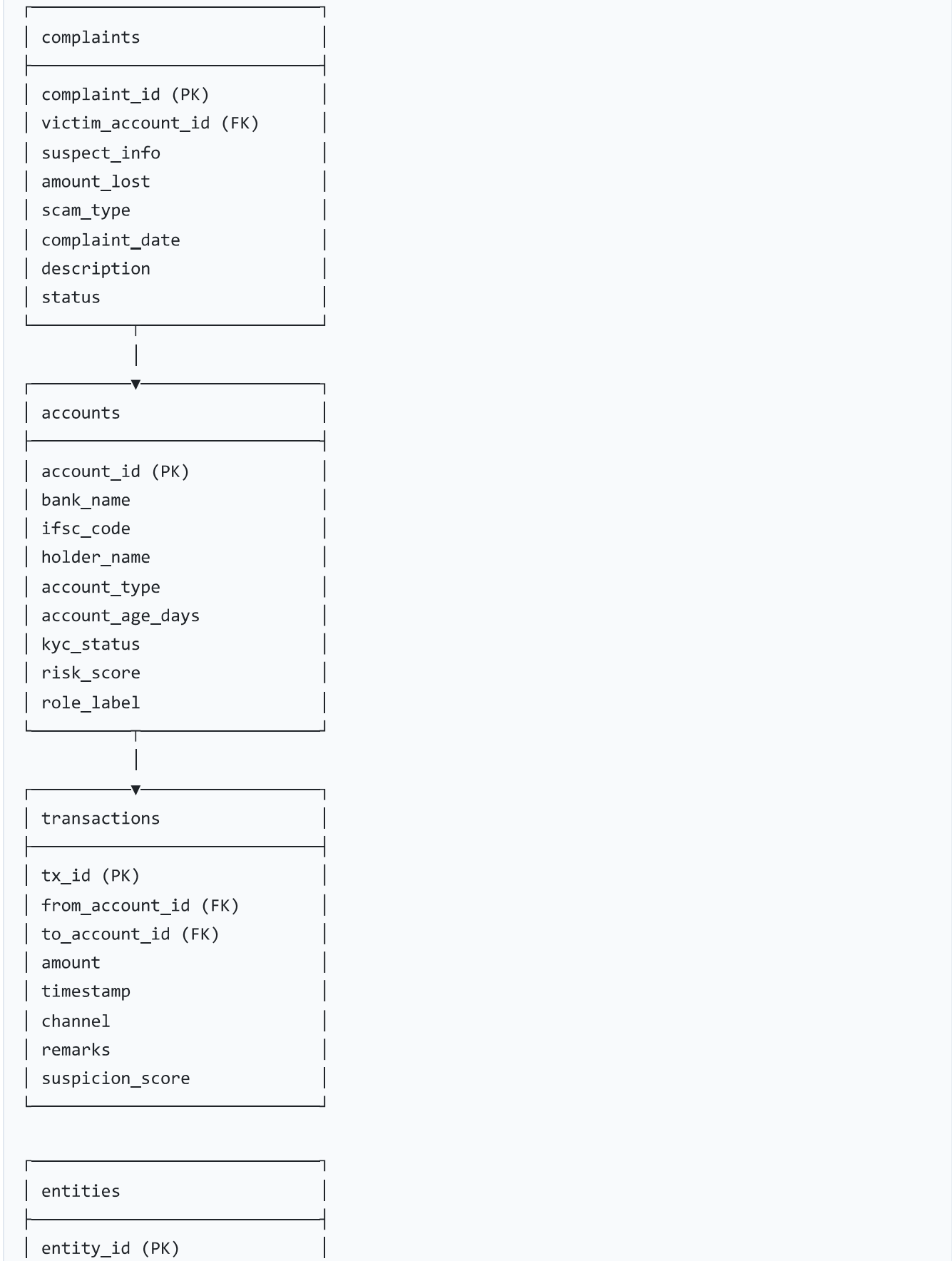
6. Data Flow Pipeline

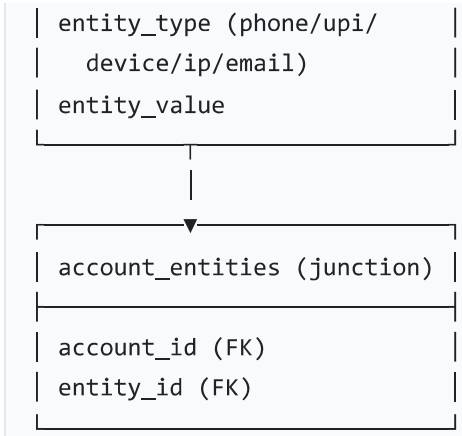
End-to-End Pipeline

7. Graph Schema & Data Model

Graph Node Types

Relational Data Model (PostgreSQL)





8. Risk Scoring Framework

Composite Risk Score Formula

$$\begin{aligned}\text{RiskScore}(node) = {} & w_1 \cdot \text{TransactionAnomaly} \\ & + w_2 \cdot \text{GraphPositionScore} \\ & + w_3 \cdot \text{ComplaintLinkageScore} \\ & + w_4 \cdot \text{FraudDNAMatchScore} \\ & + w_5 \cdot \text{TemporalBurstScore} \\ & + w_6 \cdot \text{DormancyReactivationScore}\end{aligned}$$

Score Components

Component	Weight	How It's Computed	Range
Transaction Anomaly	0.20	High inflow-outflow ratio, rapid forwarding, amount fragmentation	0–1
Graph Position	0.20	Betweenness centrality, degree centrality in suspicious subgraph	0–1
Complaint Linkage	0.20	Direct or N-hop connection to complaint-linked accounts	0–1
Fraud DNA Match	0.15	Similarity to known fraud campaign fingerprints	0–1
Temporal Burst	0.15	Volume spike vs. historical baseline (z-score normalized)	0–1
Dormancy Reactivation	0.10	Days dormant before sudden activity spike	0–1

Mule Role Classification

Based on graph features, each suspicious node is labeled:

Role	Detection Logic	Visual
Collector	First recipient from victim; high fan-in from unrelated senders	🟠 Orange
Splitter	Receives large sum, splits into 3+ smaller outgoing transfers	🔴 Red
Layering Node	Pass-through: receives and forwards ~same amount quickly	🟡 Yellow
Hub / Controller	High betweenness centrality; connects multiple mule sub-networks	⚫ Black/Purple
Cash-Out	Terminal node; ATM withdrawal or merchant payment, no onward transfer	⬛ Dark
Bridge	Appears in multiple complaint trails; connects separate fraud cases	🔵 Blue ring
Dormant Mule	Was inactive 30+ days, activated briefly for fraud, dormant again	⚪ Grey pulse

9. Explainability Engine

For Every Flagged Account

The system generates a human-readable **Risk Card**:



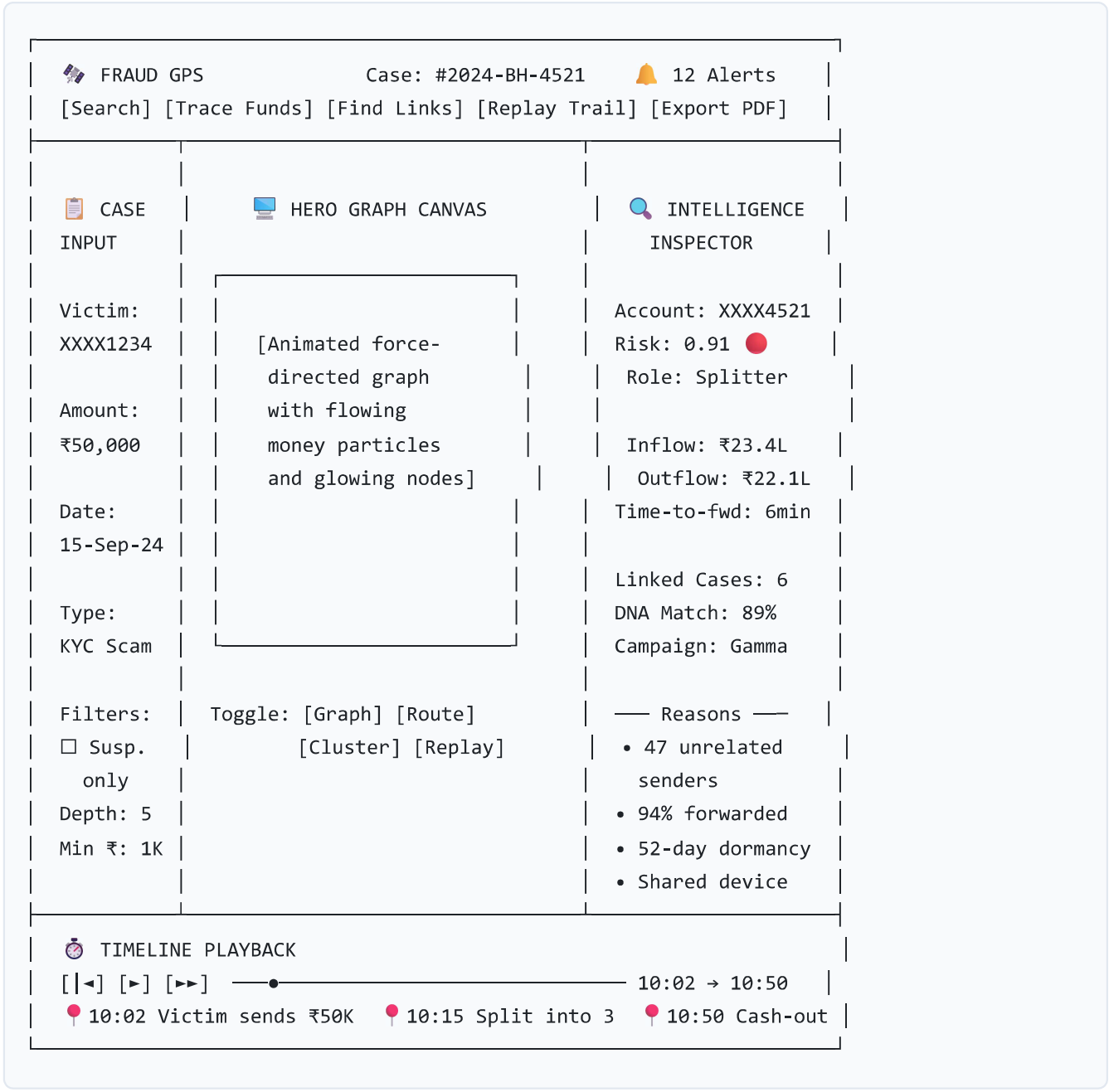
Auto-Generated Investigation Summary

One-click narrative:

> *"On 2024-09-15, victim Rajesh Kumar (Account XXXX1234, SBI Patna) transferred ₹50,000 via UPI to account XXXX4521 (PNB Varanasi) after receiving a fraudulent 'KYC expiry' call. Within 18 minutes, XXXX4521 split the funds into three transfers: ₹20,000 to XXXX7891, ₹15,000 to XXXX3456, and ₹14,500 to XXXX6789. Account XXXX7891 further split into two ATM cash withdrawals within 40 minutes. Account XXXX4521 has been identified as a SPLITTER node appearing in 6 other complaints with a combined loss of ₹12.4 lakhs. The fraud pattern matches the 'Gamma Network' campaign signature with 89% confidence. Device ID analysis links the account operator to 3 additional mule accounts."*

10. UI / UX Architecture

Screen Layout



Graph Canvas — Visual Encoding Rules

Visual Property	What It Encodes
Node color	Risk level: 🟢 Green (safe) → 🟡 Yellow → 🟠 Orange → 🔴 Red → ⚫ Critical
Node size	Transaction volume (larger = more money passed through)
Node border	Role: solid = normal, dashed = bridge, double = hub, pulsing = active alert
Node label	Masked account ID + role label
Edge thickness	Transaction amount
Edge color/brightness	Speed of transfer (brighter = faster)
Edge animation	Flowing particles showing money direction
Pulsing red edge	Flagged suspicious transfer
Dashed edge	Cross-case entity link (from Fraud DNA)
Shaded hull	Cluster boundary grouping related mule accounts

Four View Modes

Mode	What It Shows	When to Use
Graph Mode	Full force-directed network	Exploring connections
Route Mode	Highlighted path from victim to cash-out	Following the money
Cluster Mode	Grouped suspicious clusters with hulls	Identifying syndicates
Replay Mode	Step-by-step animated story	Demo & presentation

Investigation Story Mode (Key Demo Feature)

The timeline playback animates the fraud as a cinematic sequence:

```
[Step 1]  Victim node appears (blue glow, gentle pulse)
          Label: "Rajesh Kumar — ₹50,000 lost"
          ↓ (1-second pause)

[Step 2]  Animated pulse flows to first receiver
          Line grows from victim → collector
          Particles flow along the edge
          Receiver node materializes (orange)
          ↓

[Step 3]  Split animation
          Collector node pulses
          Three lines branch out simultaneously
          Money particles divide into three streams
          Each stream's thickness = amount ratio
          Three new nodes appear (yellow/red)
          ↓

[Step 4]  Layering chain lights up
          One branch continues: Mule B → Mule D → Mule E
          Sequential edge animation
          Each hop takes ~0.5 seconds
          ↓

[Step 5]  Cash-out terminals darken
          Terminal nodes turn dark/black
          "ATM Withdrawal" labels appear
          ↓

[Step 6]  Fraud DNA connections appear
          Dashed lines emerge connecting to other case nodes
          "Connected to Case #142, #308" labels fade in
          ↓

[Step 7]  Risk cards pop up
          Most suspicious node gets a glowing risk card overlay
          "SPLITTER — Risk: 0.91 — 6 linked complaints"
```

11. Tech Stack

Recommended Stack for Hackathon

Layer	Technology	Why This Choice
Frontend Framework	React + TypeScript	Fastest to build interactive UIs
Graph Visualization	Cytoscape.js	Purpose-built for network graphs; supports animations, styling, layouts
Supplementary Viz	D3.js	For Sankey money-flow diagrams and custom animations
Styling	Tailwind CSS	Rapid professional styling
UI Animations	Framer Motion	Smooth panel transitions, card reveals
Backend	Python FastAPI	Async, fast, great for data-heavy backends
Graph Analytics	NetworkX	Centrality, path finding, community detection, built into Python
ML / Scoring	scikit-learn	Clustering (DBSCAN), anomaly detection, similarity scoring
Data Processing	Pandas + NumPy	Data wrangling, feature engineering
Text Similarity	TF-IDF + cosine / FuzzyWuzzy	For Narrative DNA and beneficiary name matching
Database	PostgreSQL (or JSON files for MVP)	Reliable; JSON files are faster for hackathon
Report Generation	ReportLab / WeasyPrint	PDF export for investigation reports

TIP

Hackathon shortcut: For MVP, skip PostgreSQL entirely. Use JSON/CSV files loaded into Pandas DataFrames and NetworkX graphs in memory. This saves 2-3 hours of database setup. Switch to PostgreSQL only if the dataset grows beyond memory.

12. Demo Storyline

The 5-Minute Demo Script

This is what you show the judges. Every second counts.

[0:00 – 0:30] The Hook

> *"Every day, Bihar's cyber police receive hundreds of fraud complaints. Each officer manually traces money through bank statements — spending hours per case. Meanwhile, the same mule network is defrauding dozens of victims across the state, and nobody sees the connection. Fraud GPS changes that."*

[0:30 – 1:30] Load & Trace

- Officer enters a complaint: Victim Rajesh, ₹50,000 lost, UPI transfer on Sep 15
- Clicks **"Trace Funds"**
- The graph canvas comes alive:
  - Victim node appears (blue glow)
  - Money flows to first receiver (animated particles)
  - **Split detected** — three branches fan out
  - Layering chain lights up hop by hop
  - Cash-out terminals darken
- **"In 3 seconds, we traced what takes an officer 3 hours"**

[1:30 – 2:30] Investigate

- Officer clicks on the SPLITTER node (pulsing red)
- Right panel reveals:
  - Risk Score: 0.91
  - Role: Splitter
  - Linked to 6 complaints
  - Forwarded 94% within 30 minutes
  - 52-day dormancy before activation
- **"Every score is explained. No black box."**

[2:30 – 3:30] Fraud DNA Reveal

- Click **"Find Linked Cases"**
- Dashed lines appear connecting to other complaint networks
- Fraud DNA panel shows:
  - 89% match to "Gamma Network" campaign
  - Same device ID found in 3 other cases

    - Same behavioral pattern across 14 complaints
    - Combined losses: ₹12.4 lakhs
- **"This isn't one fraud. It's a syndicate. And Fraud DNA just exposed it."**

[3:30 – 4:30] Story Mode Replay

- Click **"Replay Trail"**
- Timeline playback animates the entire fraud step by step
- Each hop appears chronologically with timestamps
- Officers can pause at any moment to inspect
- **"Investigators can replay the crime like a surveillance recording"**

[4:30 – 5:00] Report & Impact

- Click **"Generate Report"**
- PDF downloads with complete evidence chain
- Final slide:
    - *3 seconds* to trace what takes *3 hours*
    - *14 connected cases* discovered from *1 complaint*
    - *Court-ready* evidence reports
    - **"Fraud GPS: Google Maps for stolen money."**

13. MVP vs Advanced Feature Matrix

Tier 1 — Must Have (Hackathon MVP)

Feature	Priority	Est. Hours
Synthetic data generator (transactions + complaints + entities)	🔴 Critical	3
Graph builder (NetworkX from transaction data)	🔴 Critical	2
Multi-hop money trail tracer (BFS)	🔴 Critical	3
Split / merge / rapid-forward detection	🔴 Critical	3
Risk score computation (rule-based composite)	🔴 Critical	2
Node role classification (Collector/Splitter/Hub/Cash-out)	🔴 Critical	2
Explainability text generation	🔴 Critical	2
React frontend with Cytoscape.js graph canvas	🔴 Critical	6
Animated money flow particles on edges	🔴 Critical	3
Node click → risk card panel	🔴 Critical	2
FastAPI backend with /trace, /risk, /graph endpoints	🔴 Critical	3
Subtotal		**~31 hrs**

Tier 2 — Strong Enhancement

Feature	Priority	Est. Hours
Entity DNA cross-referencing (phone, device, UPI matching)	🔴 High	3
Behavioral DNA fingerprinting + similarity scoring	🔴 High	4
Fraud DNA panel in UI	🔴 High	3
Timeline playback / Story Mode	🔴 High	4
Case linking radar (related cases panel)	🔴 High	2
Mule lifecycle labels on graph	🔴 High	1
Investigation summary auto-generation (template-based)	🔴 High	2
PDF report export	🔴 High	2
Subtotal		**~21 hrs**

Tier 3 — If Time Remains

Feature	Priority	Est. Hours
Network DNA topology comparison	🟡 Medium	4
Graph community detection (Louvain/DBSCAN)	🟡 Medium	3
Cluster hull visualization	🟡 Medium	2
Geo overlay (map of account locations)	🟡 Medium	3
Sankey money-flow diagram	🟡 Medium	3
GNN-based anomaly detection (PyTorch Geometric)	🟡 Medium	8
LLM-generated narrative summaries	🟡 Medium	4

WARNING

Do NOT start with Tier 3. Build Tier 1 completely, then add Tier 2 features. Tier 3 is bonus. A polished Tier 1 + Tier 2 demo will beat a half-finished Tier 3 attempt every time.

14. Hackathon Build Plan

Team of 4 — Role Assignment

Role	Person	Focus
Backend Lead	Person 1	FastAPI, graph algorithms, data pipeline
ML / Analytics	Person 2	Risk scoring, DNA engines, data generation
Frontend Lead	Person 3	React, Cytoscape.js, graph animations
Frontend + Report	Person 4	UI panels, timeline, PDF export, presentation

Hour-by-Hour Plan

```
PHASE 1: FOUNDATION (Hours 0–8)
═══════════════════════════════════════════════
├─ Person 1: FastAPI setup, data models, /trace endpoint
├─ Person 2: Synthetic data generator, NetworkX graph builder
├─ Person 3: React scaffold, Cytoscape.js integration, basic graph render
└─ Person 4: UI layout (sidebar + graph + inspector panel), Tailwind setup

  🎯 Milestone: Can load data and see a static graph

PHASE 2: GPS ENGINE (Hours 8–18)
═══════════════════════════════════════════════
├─ Person 1: Trail tracer (BFS), split/merge detection, /risk endpoint
├─ Person 2: Risk scoring engine, role classification, explanation generator
├─ Person 3: Animated edge particles, node color encoding, click interactions
└─ Person 4: Node inspector panel, risk card component, filters

  🎯 Milestone: Can trace money trail with animation + click for risk cards

PHASE 3: DNA ENGINE (Hours 18–30)
═══════════════════════════════════════════════
├─ Person 1: Entity DNA cross-referencing, /dna endpoint
├─ Person 2: Behavioral DNA fingerprinting, campaign clustering
├─ Person 3: Dashed cross-case links on graph, cluster hulls
└─ Person 4: Fraud DNA panel, case linking radar, timeline playback

  🎯 Milestone: Full investigation flow works end-to-end

PHASE 4: POLISH & DEMO (Hours 30–40)
═══════════════════════════════════════════════
├─ Person 1: Report generation endpoint, edge cases, error handling
├─ Person 2: Fine-tune scoring, prepare demo dataset with compelling story
├─ Person 3: Story Mode replay, visual polish, smooth animations
└─ Person 4: PDF report viewer, presentation deck, demo script practice

  🎯 Milestone: Demo-ready product

PHASE 5: FINAL PREP (Hours 40–48)
═══════════════════════════════════════════════
├─ All: End-to-end demo rehearsal (minimum 3 dry runs)
├─ All: Bug fixes, UI polish, backup demo recording
├─ All: Presentation deck finalization
└─ All: Sleep 😴 (at least a little)

  🎯 Milestone: Ready to win
```

Key Differentiators — Why This Wins

What Judges See	Why It Stands Out
Animated money trail	No other team will have a cinematic graph experience
Story Mode replay	Turns investigation into a watchable narrative
Fraud DNA cross-case intelligence	Goes beyond single-case analysis to syndicate discovery
Explainable risk cards	Every flag has clear, human-readable reasons
Role classification	Translates graph theory into police-friendly terms
Bihar-specific framing	Jamtara belt, UPI fraud, digital arrest scams
One-click report	Court-ready output that officers actually need
3 seconds vs 3 hours	Memorable impact metric for the pitch

> IMPORTANT
>
> **The single most important thing:** Make the animated graph canvas look incredible. That 10-second moment when money flows from victim through splits to cash-out terminals — that's what judges will remember. Spend disproportionate time on this.

Fraud GPS: Powered by Fraud DNA — Google Maps for stolen money.